

WASP-12b

R-0389

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230130 · created 2026-07-18T23:38:14+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459974.8032 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.103 +/- 0.0082
Transit depth (Rp/Rs)^2	1.06 +/- 0.17 [%]
Orbital Inclination (inc)	85.21 +/- 2.09
Airmass coefficient 1 (a1)	1.2012 +/- 0.0093
Airmass coefficient 2 (a2)	-0.1264 +/- 0.0037
Scatter in the residuals of the lightcurve fit is	0.78 %
Best Comparison Star	None
Optimal Aperture	4.56
Optimal Annulus	9.5
Transit Duration (day)	0.1254 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.103 ± 0.0082 vs 0.1178 (deviation 1.44σ)
Duration fit vs published	0.1254 d vs 0.125 d (deviation 0.11σ)
Mid-time O–C	-3.4 min
Depth SNR	10.0
Residual scatter	0.78 %

Light curve

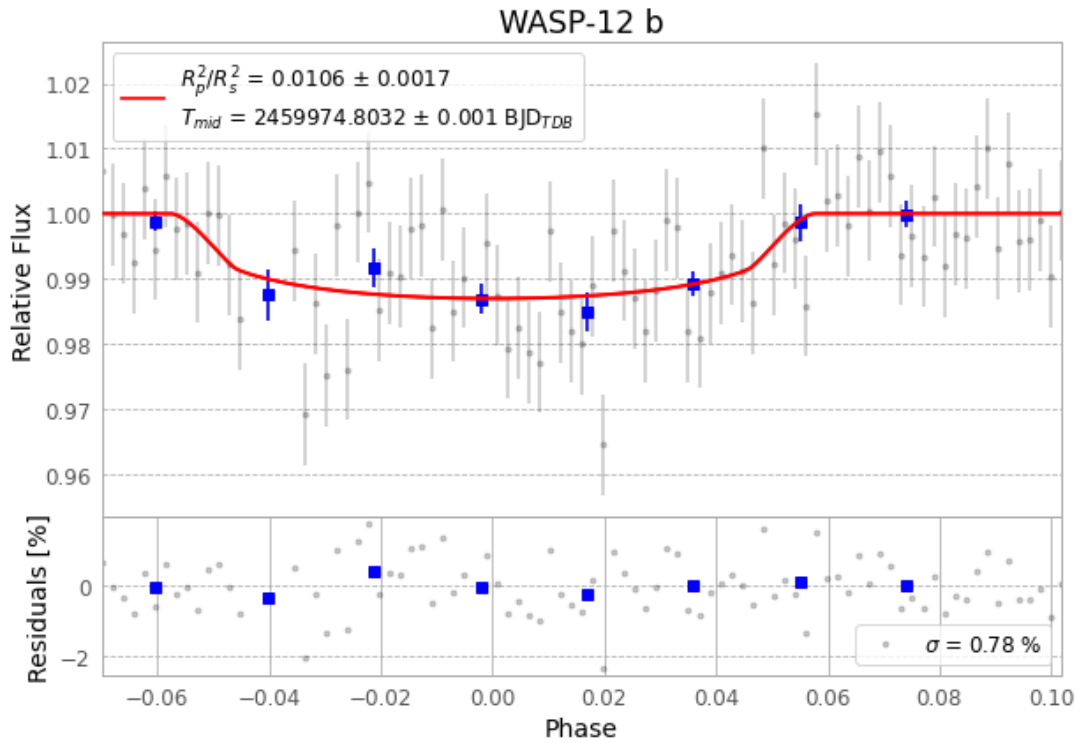


Figure 1 — FinalLightCurve_WASP-12 b_2023-01-29.png (SHA-256 d031e7ebc98a06e4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 222], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 253968bed335b0c9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

91 FITS frames (60 MB), WASP-12230130051815.FITS → WASP-12230130094815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230130.log (d96cf767b598...), FinalLightCurve_WASP-12 b_2023-01-29.png (d031e7ebc98a...), FinalLightCurve_WASP-12 b_2023-01-29.csv (62a1a21d016f...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 23:38Z.